Balanced envy-freeness up to one good for five agents and the chromatic number of a balanced generalized Kneser graph

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Abstract

Two fixed groups of agents share indivisible goods, with one bundle assigned to each group. Each agent assigns a value to every bundle. Valuations are monotone: adding goods never reduces value. A balanced allocation has equal bundle sizes. It is envy-free up to one good (EF1) if each agent values her group's bundle at least as highly as the other bundle after removing at most one good.

We prove that at most five agents always admit a balanced EF1 allocation when the number of goods is even; for odd numbers, the bundle sizes differ by one. The bound is sharp for every split of six agents. These conclusions follow from a known link between group fair division and graph coloring, together with a new theorem resolving a 2017 conjecture. The graph has the half-size subsets of an even ground set as vertices; adjacent subsets share at most one element. Its chromatic number, the minimum number of colors separating adjacent vertices, is six for ground sets of at least four elements. For larger ground sets, the standard topological coloring bound gives only five colors. We obtain the sixth by describing three-dimensional cycles modulo boundaries: local coloring constraints force a hypothetical five-coloring to send all these classes to zero, while symmetry forces a nonzero image.

For even numbers of goods, the same six-color theorem gives balanced envy-freeness up to two goods (EF2) for six agents when one valuation sums nonnegative item weights (additive) and the others are monotone. We also construct a balancing map and give polynomial-time reductions between finding a fair allocation and finding equally colored adjacent vertices for fixed removal allowances in the guaranteed range. Explicit colorings provide counterexamples for larger allowances.

Keywords. Fair division; group allocation; approximate envy-freeness; generalized Kneser graphs; chromatic number; topological combinatorics; total search.

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1 Introduction

Setting. When goods are shared between two families, teams, or departments, each group receives one bundle but its members need not agree on its value. We fix the two groups in advance and require fairness for every member. A valuation assigns a nonnegative real value to each bundle. It is monotone if adding goods never reduces this value. It is additive if each good has a nonnegative weight and a bundle's value is the sum of its item weights. For a nonnegative integer c, an allocation is envy-free up to c goods, or EFc, if each agent values her group's bundle at least as highly as the other bundle after removing at most c goods from the latter. The removed goods may depend on the agent. EF1 is the case $c = 1$, and EF2 is

the case $c = 2$. A balanced allocation has equal bundle cardinalities. When the total number of goods is odd, we require an almost-balanced allocation, whose bundle sizes differ by one. This is a cardinality constraint rather than a comparison of values; such constraints arise naturally in fair division [5].

Our model has two shared bundles, regardless of the number of agents. Individual fair division instead gives each agent a separate bundle. The five-agent theorem here concerns two group bundles.

Main results. For positive integers N, k, d with $1 \le d \le k \le N$, write $[N] = \{1, \dots, N\}$. The generalized Kneser graph $\mathrm{KG}(N, k, d)$ has all k-element subsets of $[N]$ as vertices; two distinct vertices are adjacent when their intersection has fewer than d elements. A proper coloring assigns colors to vertices so that adjacent vertices have different colors. For a graph G, its chromatic number $\chi(G)$ is the smallest number of colors in a proper coloring. For a finite set M and an integer $r \ge 0$, write $\binom{M}{r} = \{A \subseteq M : |A| = r\}$; with integer arguments, $\binom{N}{r}$ is the usual binomial coefficient. We write 2^M for the family of all subsets of M, and A^c for the complement of A within its current ground set. All parameters counting vertices, goods, colors, or dimensions are integers.

For an integer $n \ge 2$, put $G_n = \mathrm{KG}(2n, n, 2)$. Its vertex set is the middle layer: the family of all subsets containing half the ground-set elements. Adjacent sets share at most one element. The 2017 conjecture [1] asserts that this graph has chromatic number six. Its consequence for fair division was identified in [19, Corollary 4.5 and Conjecture 4.6]: balanced EF1 for two fixed groups containing at most five agents in total, with arbitrary monotone valuations. We prove the conjecture and obtain that guarantee.

Theorem 1.1. *For every integer $n \ge 2$,*

$$\chi(\mathrm{KG}(2n, n, 2)) = 6.$$

Corollary 1.2. *Every two-group instance with at most five agents and arbitrary monotone valuations admits a balanced EF1 allocation when the number of goods is even, and an almost-balanced one when it is odd. For every prescribed split of six agents and every even number of goods at least four, there are monotone valuations for which no balanced EF1 allocation exists.*

Writing (a, b) for the two group sizes, the corollary resolves the conditional cases $(1, 4)$ and $(2, 3)$ and their reversals. Its sharpness for every split follows from the chromatic characterization in [19]; we also give a simple explicit family of six-agent counterexamples.

For even numbers of goods, the same six-color theorem gives balanced EF2 for six agents when one valuation is additive and the others are monotone (Corollary 6.3). Pairing the goods reduces the allocation space to choices of one good from each pair, and a projection of a Kneser graph transfers the six-color lower bound to these choices. Thus the EF1 and EF2 guarantees arise from one coloring theorem.

The sixth color. The neighborhood complex of a graph consists of the vertex families that have a common neighbor, with such families serving as its faces. These families are closed under taking subsets, which is the defining property of a simplicial complex. Lovász's method bounds the chromatic number through the connectivity of this complex [20]. For G_n, a family has a common neighbor precisely when each set can be obtained from one central n-set by replacing at most one element. The collection of all such sets is called a Johnson ball of radius one. This complex is 2-connected: it is path-connected, and every continuous map from a circle or a two-dimensional sphere extends to the corresponding ball. Its third

homology, which records three-dimensional cycles modulo boundaries, is nonzero for $n \geq 3$. The connectivity bound therefore stops at five for $n \geq 3$.

We seek the sixth color in the homology that remains. For $1 \leq k < N$, the hypersimplex is the polytope $\Delta(N,k) = \{x \in [0,1]^N : \sum_{i=1}^{N} x_i = k\}$. Intersecting $\Delta(2n,n)$ with linear half-spaces gives closed convex pieces called caps. The union of the caps chosen in the proof gives a geometric model for the complex and can be continuously deformed onto the faces of dimension at most three together with all simplex faces. The boundaries of the missing four-faces, copies of $\Delta(5,2)$ and $\Delta(5,3)$, generate its third homology. Each generator is represented by a boundary cycle on five n-sets arranged in a pentagon. These sets vary on five coordinates, called their free coordinates, while the other coordinates are fixed. A hypothetical five-coloring restricts locally to an edge-coloring of the complete graph on five vertices, denoted K_5. The pentagon lemma prevents a rainbow pentagon, whose five edges all have different colors, forcing the induced map to annihilate every generator. An argument using the symmetry that interchanges opposite points on a sphere forces that same map to be nonzero. The contradiction excludes five colors without requiring an additional degree of connectivity.

Further results. For an agent, the rejected allocations are those that fail EFc for that agent. No two rejected allocations are adjacent in the appropriate graph; this is the meaning of independence used in the argument of [19], which identifies the sharp balanced EFc threshold on an even number $2r$ of goods, for $r \geq c+1$, as $\chi(\mathrm{KG}(2r,r,c+1)) - 1$. We recall its converse construction because it also underlies our search reductions. The known lower bound $\chi(\mathrm{KG}(2r,r,c+1)) \geq 2c+2$ [7] therefore guarantees balanced EFc for at most $2c+1$ agents; if $r \leq c$, the entire comparison bundle may be deleted. For an integer $m \geq 2c+1$, identify an allocation with the subset of $[m]$ assigned to the first group. The full allocation graph $\Gamma_{c,c}(m)$ has all these subsets as vertices; adjacent allocations have at most c common chosen goods and at most c common unchosen goods. A complement-preserving balancing map proves

$$\chi(\Gamma_{c,c}(m)) = \chi\big(\mathrm{KG}(2\lceil m/2\rceil, \lceil m/2\rceil, c+1)\big), \qquad m \geq 2c+1.$$

Balance thus preserves the existence guarantee supplied by the chromatic method.

A circuit can specify a coloring or valuation on exponentially many subsets without listing all its values. Under this succinct representation, the EFc search task asks for a fair allocation. If a valuation circuit fails monotonicity, the solver may instead identify that valuation u and report bundles $S \subseteq T$ with $u(S) > u(T)$. When the number of agents is below $\chi(\mathrm{KG}(2r,r,c+1))$, this makes the task total: every encoded input has a permitted solution. For fixed c and a prescribed group split, it is polynomial-time many-one equivalent to finding a monochromatic edge—two adjacent vertices assigned the same color—in a coloring of $\mathrm{KG}(2r,r,c+1)$, whenever the number of colors is below the chromatic number. Here each reduction constructs a single instance of the other problem and decodes any of its solutions. If L is the input description length, the reductions take $L^{O(1)} r^{O(c)}$ time. The exponent of r depends on the removal allowance c, so the bound is polynomial for fixed c. In particular, five-color search is total, while finding a polynomial-time algorithm for it remains open. Its membership in Polynomial Parity Argument (PPA), the class of total-search problems reducible to finding a second odd-degree vertex when one is given in a succinctly specified undirected graph, also remains open. For larger removal allowances, explicit colorings yield monotone counterexamples with 14, 30, 62, and 126 agents for EF2 through EF5, respectively, in the ranges stated in Corollary 7.6.

Related work. The classical Kneser graph is the case $d = 1$ [17, 20]. General intersection-threshold graphs were studied in [10]; the conjecture settled here concerns the middle layer

with intersection threshold two. The graph-theoretic approach to balanced group fair division was developed in [19]. A weakly polynomial-time algorithm for four agents with additive valuations, with running time polynomial in the bit lengths of the item weights, appears in [11]; monotone valuations and allocator preferences on the full allocation space are studied in [6]. The five-agent balanced existence theorem resolves the remaining conditional cases, and the search equivalence supplies their computational formulation. It belongs alongside the original Kneser search problem [9], related graph-search results [13], and the connection between Kneser search and resource allocation [14]. The upper colorings use families forcing a large two-color imbalance on at least one prescribed set, a discrepancy construction [3]. Related methods use Hadamard matrices, square matrices with entries 1 and -1 and mutually orthogonal rows [16].

Organization. Section 2 proves the coloring theorem. Sections 3 and 4 give the fair-division thresholds and the balancing map. Section 5 establishes the search equivalence, Section 6 proves the additive-agent transfer, and Section 7 gives the explicit obstructions.

2 The six-color theorem

Throughout this section, $G_n = \mathrm{KG}(2n, n, 2)$ and $n \geq 2$. We use a homological refinement of the neighborhood-complex bound. All graphs below are finite and have no loops. For a graph G, write $V(G)$ and $E(G)$ for its vertex and edge sets, K_q for the complete graph on q vertices, and $v \sim w$ for adjacency. A graph homomorphism sends vertices to vertices and edges to edges; $G \to H$ means that one exists. It follows that $\chi(G) \leq \chi(H)$. A proper q-coloring is equivalently a homomorphism to K_q.

For a finite nonempty vertex set W, the simplex $\Delta(W)$ consists of probability vectors on W. A face with $j+1$ vertices has dimension j; Δ^j denotes a simplex of dimension j, and $\partial\Delta^j$ its boundary. For a simplicial complex K, its geometric realization $|K|$ consists of formal convex combinations $\sum_v t_v v$, where the coefficients are nonnegative, sum to one, and have positive support in a face of K. Its j-skeleton is the union of its faces of dimension at most j. The one-argument simplex notation $\Delta(W)$ is distinct from the two-argument hypersimplex notation $\Delta(N, k)$ introduced above.

We use homology over the two-element field $\mathbb{F}_2 = \mathbb{F}_2$. For a space X, write $H_j(X; \mathbb{F}_2)$ for its jth homology group and $\pi_j(X)$ for its jth homotopy group. For a complex, these refer to its geometric realization. A nonempty space is a-connected, for $a \geq 0$, if it is path-connected and $\pi_j = 0$ for $1 \leq j \leq a$. The sphere S^j is the unit sphere in $\mathbb{R}^{j+1}$. A homotopy continuously deforms one map into another. A homotopy equivalence has an inverse up to such deformations; a contractible space can be continuously deformed to a point. A simplicial map sends vertices to vertices and each face to a face; $|f|$ denotes its continuous map between geometric realizations. For a continuous or simplicial map f, the symbol f_* denotes its induced map on homology.

The neighborhood complex $\mathcal{N}(G)$ consists of vertex families with a common graph neighbor. A proper coloring $\kappa : G \to K_q$ sends each such face to a set of colors missing the color of a common neighbor. It therefore induces a simplicial map $\kappa_{\mathcal{N}} : \mathcal{N}(G) \to \mathcal{N}(K_q)$, defined by applying κ to every vertex of the face. The following criterion applies to these particular maps.

Proposition 2.1 (Generator-annihilation criterion)**.** *Let $t \geq 1$, and let G be a finite loopless graph with $(t-1)$-connected neighborhood complex. Suppose that classes $\{\gamma_\alpha\}$ generate $H_t(\mathcal{N}(G); \mathbb{F}_2)$ and that every proper coloring $\kappa : G \to K_{t+2}$ satisfies $(\kappa_{\mathcal{N}})_* \gamma_\alpha = 0$ for all α. Then $\chi(G) \geq t + 3$.*

Proof. Such a coloring would induce the zero map on H_t, contrary to Lemma 2.8. A coloring with fewer than $t+2$ colors can be composed with the inclusion into K_{t+2}, so it is excluded as well. □

For G_n, the proof proceeds from geometry to coloring. Lemma 2.3 classifies the cap intersections containing no point whose coordinates are all zero or one. Proposition 2.4 shows that the affine map to the cap union preserves third homology. Lemma 2.5 deforms that union onto the three-skeleton together with all simplex faces. Theorem 2.6 then represents its third-homology generators by pentagon cycles associated with the omitted four-faces. Proposition 2.7 uses local edge-coloring constraints to show that every five-coloring annihilates these generators. Lemma 2.8 forces the same coloring map to be nonzero on third homology, producing the contradiction.

Here is a direct six-coloring for every $n \geq 2$. Fix a three-element set T. Each n-set A either contains a pair $p \subset T$ or avoids a pair $p \subset T$. Give A one available label $(+, p)$ or $(-, p)$, with a fixed rule to break ties. Two sets with the same positive label meet in p. Two with the same negative label lie in a common $(2n-2)$-element set, and hence meet in at least two elements. The six labels therefore give a proper coloring. The same upper bound also follows from [1, Theorem 2.2].

It remains to exclude five colors. We use homology over $\mathbb{F}_2$. The obstruction has two parts: topology forces a five-coloring to induce a nonzero map on third homology, while a local pentagon lemma forces this particular map to be zero. The geometric work is to exhibit generators on which the local lemma applies.

2.1 A local pentagon obstruction

For a pair $p = \{u, v\}$ of vertices of K_5, write

$$I(p) = \{e \in E(K_5) : e \cap p \neq \varnothing\}.$$

A pair is a two-vertex cover of an edge family when it meets every edge. The family has such a cover exactly when it is contained in some $I(p)$. A five-cycle is rainbow when its five edges receive distinct colors.

Lemma 2.2 (No rainbow pentagon)**.** *Color the edges of K_5 with at most five colors. If each $I(p)$ uses at most four colors, then no Hamiltonian five-cycle is rainbow.*

Proof. Suppose a rainbow cycle exists. Index its vertices by $\mathbb{Z}/5\mathbb{Z}$, put $e_i = \{i, i+1\}$, and write κ for the edge-coloring, naming its colors so that $\kappa(e_i) = i$. Write $d_i = \{i, i+2\}$ for the chords. All subscripts in this proof are modulo five.

Replacing e_j by d_i gives a five-edge family with a two-vertex cover for $j = i, i+1, i+3$. The respective covers are

$$\{i+2, i+4\}, \qquad \{i, i+3\}, \qquad \{i, i+2\}.$$

Thus $\kappa(d_i)$ cannot equal $i, i+1$, or $i+3$, since that replacement would be a rainbow family with a two-vertex cover. Consequently

$$\kappa(d_i) \in \{i+2, i-1\}. \tag{1}$$

Set $a_i = 0$ for the first choice in (1), and $a_i = 1$ for the second.

The cycle edges in $I(\{i, i+1\})$ have colors $i-1, i, i+1$. The two remaining colors occur among its chords as follows:

$$\begin{aligned} i+2 \text{ occurs} &\iff a_i = 0 \text{ or } a_{i-2} = 1, \\ i+3 \text{ occurs} &\iff a_{i+1} = 0 \text{ or } a_{i-1} = 1. \end{aligned}$$

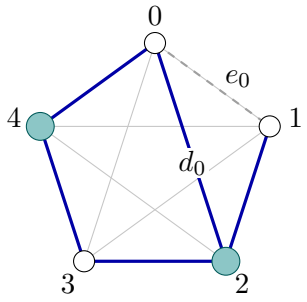


Figure 1: Replacing e_0 by the chord d_0 gives five edges covered by the vertices 2 and 4. The omitted edge is dashed. This is the first of the three replacements used in Lemma 2.2.

At least one must be absent, so

$$(a_i = 1 \text{ and } a_{i-2} = 0) \quad \text{or} \quad (a_{i+1} = 1 \text{ and } a_{i-1} = 0). \tag{2}$$

Let $B = \{i : a_i = 1,\ a_{i-2} = 0\}$. Equation (2) says that B meets every adjacent pair of the ordinary five-cycle, whence $|B| \geq 3$. But B cannot contain both i and $i+2$: these would require $a_i = 1$ and $a_i = 0$. Thus B is independent in the five-cycle whose steps are two, and $|B| \leq 2$. This is a contradiction. □

2.2 The Johnson complex and a hypersimplex

Fix $2 \leq k \leq N-2$. The Johnson distance on $\binom{[N]}{k}$ is

$$d_J(A, B) = k - |A \cap B|.$$

The Johnson graph $J(N, k)$ has these k-sets as vertices and joins the pairs at distance one.

Define $B_1^J(Z) = \{A \in \binom{[N]}{k} : d_J(A, Z) \leq 1\}$. The radius-one ball complex (or discrete Čech complex) of the Johnson graph is $\mathcal{C} = \mathcal{C}(N, k)$, the simplicial complex whose nonempty faces are the families contained in a common closed Johnson ball of radius one. Thus $\sigma \in \mathcal{C}$ if some k-set Z satisfies $d_J(A, Z) \leq 1$ for all $A \in \sigma$.

For a vector $x = (x_1, \ldots, x_N) \in \mathbb{R}^N$ and any $I \subseteq [N]$, write $x(I) = \sum_{i \in I} x_i$. Consider the hypersimplex and its caps

$$P = \Delta(N, k) = \left\{ x \in [0,1]^N : \sum_i x_i = k \right\}, \qquad P_A = \{x \in P : x(A) \geq k - 1\}.$$

For an indexed family $\mathcal{V} = (V_i)_{i \in I}$, its nerve $\mathrm{Nrv}(\mathcal{V})$ is the complex of finite nonempty index sets $J \subseteq I$ with $\bigcap_{i \in J} V_i \neq \varnothing$. Put $U = \bigcup_A P_A$ and $Y = \mathrm{Nrv}((P_A)_A)$. The indicator vector $\mathbf{1}_B \in \{0,1\}^N$ has coordinate one at elements of B and zero elsewhere. It lies in P_A exactly when $d_J(A, B) \leq 1$, giving $\mathcal{C} \subseteq Y$.

For a point $\sum_A t_A A$ of $|\mathcal{C}|$, the coefficients t_A are its nonnegative simplex weights. There is a continuous map

$$f : |\mathcal{C}| \longrightarrow U, \qquad \sum_A t_A A \longmapsto \sum_A t_A \mathbf{1}_A. \tag{3}$$

Indeed, a simplex with common Johnson center Z maps into the convex cap P_Z. Proposition 2.4 identifies the homology map induced by f.

Exceptional nerve faces.

Lemma 2.3. *The faces of Y absent from $\mathcal{C}$ are precisely*

$$\mathcal{O}(R,S) = \{R \cup e : e \in \tbinom{S}{2}\}, \qquad |R| = k-2, \quad |S| = 4, \quad R \cap S = \varnothing. \tag{4}$$

Each is a maximal five-simplex of Y, and its whole boundary belongs to $\mathcal{C}$. The six centers, joined at Johnson distance one, form the octahedral graph $J(4,2)$; their indicator convex hull is the three-dimensional octahedron $\Delta(4,2)$, whereas the nerve simplex has dimension five.

Proof. For k-sets A, B at distance d and any $x \in P$,

$$x(A) + x(B) = x(A \cup B) + x(A \cap B) \le k + (k-d) = 2k - d. \tag{5}$$

Thus a nonempty intersection $P_A \cap P_B$ requires $d \le 2$. In a family of caps with nonempty intersection, if all centers have pairwise distance at most one, any center is already a discrete common center. Otherwise choose a distance-two pair

$$A = R \cup \{a,b\}, \qquad B = R \cup \{c,d\}, \qquad |R| = k-2.$$

Equality in (5) forces

$$\begin{aligned} x_i &= 1 && (i \in R), \\ x_i &= 0 && (i \in [N] \setminus (R \cup \{a,b,c,d\})), \\ x_a + x_b &= x_c + x_d = 1. \end{aligned}$$

Consequently $P_A \cap P_B$ is exactly a square. Each corner is the indicator vector $\mathbf{1}_Z$ of a k-set Z; if it lies in all the caps under consideration, that Z is a discrete common center. All coordinates outside $S = \{a,b,c,d\}$ are now fixed; those indexed by S are the free coordinates. The square coordinates are

$$x_a = u, \quad x_b = 1-u, \quad x_c = v, \quad x_d = 1-v, \qquad 0 \le u, v \le 1.$$

We classify the restriction of the cap P_D indexed by an additional k-set D. Write

$$S = \{a,b,c,d\}, \qquad \delta = (k-2) - |D \cap R|, \qquad T = D \cap S.$$

Since $|D| = k$, we have $|T| \le \delta + 2$, and its inequality on the square is

$$\sum_{i \in T} x_i \ge 1 + \delta. \tag{6}$$

For $\delta = 0$, T contains at most two of the free coordinates. Taking the pair $\{a,b\}$ or $\{c,d\}$ gives the whole square; taking one coordinate gives a side; taking none gives the empty set. A cross-pair gives one of the four half-squares

$$u + v \ge 1, \qquad u + v \le 1, \qquad u \ge v, \qquad u \le v. \tag{7}$$

For $\delta = 1$, the required sum is at least two. If $|T| = 3$, the omitted coordinate is forced to be zero, giving a side. If $|T| = 2$, the condition is either impossible or forces a corner. Fewer coordinates are impossible. For $\delta \ge 2$, the required sum exceeds the total free mass, which is two, so the condition is impossible.

A nonempty intersection containing a side or corner condition contains a square corner. To check the side case, restrict the remaining inequalities to that side: each is vacuous, impossible, or forces an endpoint. Without such conditions, only (7) matter. Each excludes exactly one square corner. Hence an intersection is nonempty but contains no corner precisely when all four diagonal inequalities are imposed. Their common intersection is $(1/2, 1/2)$.

The four diagonal inequalities arise from the four cross-pair centers. Together with A and B they give exactly (4). At the square center, a containing cap must have $\delta = 0$ and $|T| = 2$. Thus exactly these six caps contain the point, proving maximality. Finally, if the center $R \cup e$ is omitted, then $R \cup (S \setminus e)$ is within Johnson distance one of every remaining center. Thus every proper subfamily belongs to $\mathcal{C}$. □

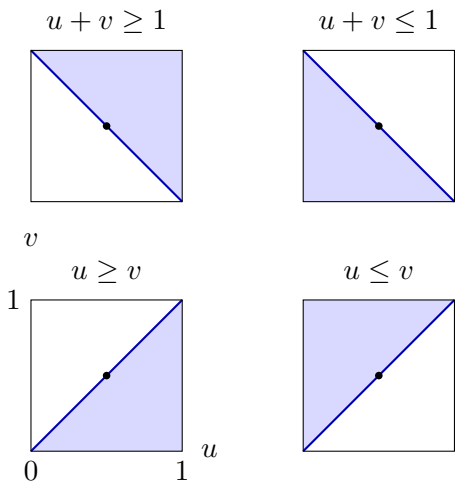


Figure 2: The four diagonal half-squares in $P_A \cap P_B$. Each contains three corners; their common intersection is the marked point $(1/2, 1/2)$. This is the exceptional corner-free intersection in Lemma 2.3.

The affine map.

Proposition 2.4. *The map f in* (3) *induces isomorphisms on $H_3(-;\mathbb{F}_2)$ and on π_i for $1 \leq i \leq 3$.*

Proof. Realize $|\mathcal{C}|$ in the standard simplex on its vertex set and cover it by

$$S_A = \Delta(B_1^J(A)).$$

Each S_A is a closed convex simplex in this realization. Its nonempty intersections are simplexes. By symmetry of Johnson distance, the nerve of this cover is $\mathcal{C}$ again: a family of indices has a common vertex in its balls exactly when the indices lie in a common ball. Also $f(S_A) \subseteq P_A$.

We compare the covers in a way compatible with f. For a finite closed convex cover $\mathcal{V} = (V_i)$, define its blowup space by

$$\mathrm{Bl}(\mathcal{V}) = \bigcup_{\varnothing \neq J \in \mathrm{Nrv}(\mathcal{V})} \left(\bigcap_{i \in J} V_i \right) \times \Delta(J).$$

The projections $\alpha_{\mathcal{V}}(x,t) = x$ to the covered space and $\beta_{\mathcal{V}}(x,t) = t$ to the nerve are homotopy equivalences by the finite closed-convex nerve theorem [4, Theorem 3.9]. Apply this to $\mathcal{S} = (S_A)_A$ and $\mathcal{P} = (P_A)_A$. Let $\iota : |\mathcal{C}| \to |Y|$ be inclusion. The map $\widehat{f}(x,t) = (f(x),t)$ gives the strictly commuting diagram

$$\begin{array}{ccccc} |\mathcal{C}| & \xleftarrow{\alpha_{\mathcal{S}}} & \mathrm{Bl}(\mathcal{S}) & \xrightarrow{\beta_{\mathcal{S}}} & |\mathcal{C}| \\ \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle \widehat{f}} & & \downarrow{\scriptstyle \iota} \\ U & \xleftarrow[\alpha_{\mathcal{P}}]{} & \mathrm{Bl}(\mathcal{P}) & \xrightarrow[\beta_{\mathcal{P}}]{} & |Y| \end{array} \tag{8}$$

All four horizontal maps are homotopy equivalences.

By Lemma 2.3, Y is obtained from $\mathcal{C}$ by attaching five-cells along their complete four-sphere boundaries. Hence ι is an isomorphism on H_3 and on π_i for $i \leq 3$, by relative cellular approximation. Diagram (8) gives the same conclusion for f. In particular, on third homology,

$$f_* = (\alpha_{\mathcal{P}})_* (\beta_{\mathcal{P}})_*^{-1} \iota_* (\beta_{\mathcal{S}})_* (\alpha_{\mathcal{S}})_*^{-1}.$$

□

Radial retraction. Let $P^{(3)}$ be the union of the faces of P of dimension at most three, and set

$$Q = P^{(3)} \cup \{\text{all simplex faces of } P\}.$$

This is a face subcomplex of P.

A strong deformation retraction of U onto Q is a continuous deformation from the identity on U to a map into Q that fixes every point of Q throughout.

Lemma 2.5. *The space U strongly deformation retracts onto Q. It is 2-connected, and $H_3(U;\mathbb{F}_2)$ is generated, under this retraction, by the boundaries of the non-simplex four-dimensional faces of P.*

Proof. A positive-dimensional face F of P is obtained by fixing some coordinates to one and some to zero, leaving m free coordinates with sum ℓ, where $1 \le \ell \le m-1$. Thus $F \cong \Delta(m,\ell)$ and $\dim F = m-1$. Its barycenter b_F has all free coordinates equal to ℓ/m.

At any $x \in F$, choose all fixed-one coordinates and the ℓ largest free coordinates. Their sum is at least $k - \ell + \ell^2/m$, with equality at b_F. Therefore

$$k - \max_{|A|=k} b_F(A) = \frac{\ell(m-\ell)}{m}. \tag{9}$$

For a simplex face, $\ell = 1$ or $m-1$, and this expression is less than one. For $m \le 4$, it is at most one. The same lower bound for the sum of the largest free coordinates holds at every point of F, so $Q \subseteq U$.

If F is non-simplex and has dimension at least four, then $m \ge 5$ and $2 \le \ell \le m-2$. Hence

$$\ell(m-\ell) \ge 2(m-2) > m.$$

By (9), b_F is outside every cap.

For $x \in U \cap F$, let $\lambda_F(x) \ge 1$ be the factor at which the ray from b_F through x first meets ∂F. Radial projection is the endpoint of the homotopy

$$H_F(x,t) = b_F + \big(1 + t(\lambda_F(x) - 1)\big)(x - b_F), \qquad 0 \le t \le 1. \tag{10}$$

This is continuous on $U \cap F$, since the excluded center does not lie there, and it fixes $U \cap \partial F$. Moreover it stays in U. Indeed, if $x(A) \ge k - 1 > b_F(A)$ and $\lambda \ge 1$, then

$$\big(b_F + \lambda(x - b_F)\big)(A) = x(A) + (\lambda - 1)(x(A) - b_F(A)) \ge k - 1.$$

Thus each point stays in every cap that contained it initially, until it reaches the face boundary.

Apply these homotopies to the omitted faces in decreasing dimension. At each stage, extend by the identity on the other remaining faces. The extensions agree on intersections because (10) fixes boundaries. There is no retained higher-dimensional simplex containing an omitted face: all faces of a simplex are simplexes. The finite pasting lemma therefore gives a strong deformation retraction $\mathcal{R} : U \to Q$.

The relative cellular chain group $C_j(P,Q)$ over $\mathbb{F}_2$ has one basis vector for each j-dimensional face of P outside Q; its boundary keeps only facets outside Q. Its homology is denoted $H_j(P,Q;\mathbb{F}_2)$. The pair (P,Q) has no such relative cells of dimension at most three. Since P is contractible, relative cellular approximation implies that Q is path-connected with $\pi_1(Q) = \pi_2(Q) = 0$. The homology exact sequence also gives

$$H_4(P,Q;\mathbb{F}_2) \xrightarrow{\cong} H_3(Q;\mathbb{F}_2). \tag{11}$$

Every relative four-cell is a cycle, because $C_3(P,Q) = 0$. Thus the boundaries of the omitted four-cells generate $H_3(Q)$. They are exactly the faces with $m = 5$ and $\ell = 2$ or 3, that is, copies of $\Delta(5,2)$ and $\Delta(5,3)$. □

Pentagon cycles. Fix a five-element set $S = \{s_0, \dots, s_4\}$ with a cyclic order. With indices modulo five, write $e_i = \{s_i, s_{i+1}\}$. In each construction, R is the fixed part, called the core, and is disjoint from S. There are two types of five vertices of $\mathcal{C}$:

$$A_i = R \cup e_i \quad (|R| = k-2), \qquad \text{or} \qquad A_i = R \cup (S \setminus e_i) \quad (|R| = k-3), \tag{12}$$

where $R \cap S = \varnothing$. The second type requires $k \geq 3$. Each type is used when a face with the indicated core and free coordinates exists.

Every four of these vertices form a face of $\mathcal{C}$. After omitting e_j, the remaining four cycle edges have the two-vertex cover $p = \{s_{j+2}, s_{j+4}\}$. A common Johnson center is $R \cup p$ in the first case and $R \cup (S \setminus p)$ in the second. Brackets below denote simplex chains over $\mathbb{F}_2$, and a hat means that a vertex is omitted. The boundary of a simplex is the sum of its codimension-one faces. Therefore

$$z(R, S) = \sum_{j=0}^{4} [A_0, \dots, \widehat{A_j}, \dots, A_4] \tag{13}$$

is a simplicial three-cycle over $\mathbb{F}_2$.

Theorem 2.6 (Pentagon homology of the Johnson ball complex)**.** *For $2 \leq k \leq N-2$, the complex $\mathcal{C}(N,k)$ is 2-connected. For each non-simplex four-face of P, choose any one Hamiltonian five-cycle on its free coordinates and form* (13)*, using its appropriate type. These chosen cycles generate $H_3(\mathcal{C}(N,k); \mathbb{F}_2)$.*

Proof. The one-skeleton contains the connected Johnson graph, so $\mathcal{C}(N,k)$ is path-connected. Proposition 2.4 and Lemma 2.5 then give $\pi_1 = \pi_2 = 0$. To identify generators, take a face $F \cong \Delta(5,2)$. The five edge-indicator vectors $\mathbf{1}_{e_i}$ are affinely independent. In fact a linear dependence of these five columns gives $a_{i-1} + a_i = 0$ at each coordinate, and the odd cycle forces every real coefficient to vanish. Their average is $(2/5, \dots, 2/5)$, the barycenter of F.

Their convex hull is consequently a full-dimensional four-simplex $T_F \subseteq F$ with b_F in its interior. Its boundary is exactly the affine image under f of the pentagon cycle. Each tetrahedral facet of this boundary lies in a cap, by the common-center argument above, so $\partial T_F \subset U$. Write $[\partial F]$ for the nonzero mod-two homology class of the three-sphere ∂F. Radial projection from b_F is a homeomorphism $\partial T_F \to \partial F$ and sends its fundamental class to $[\partial F]$ over $\mathbb{F}_2$. Complementing the free coordinates gives the same statement for $\Delta(5,3)$.

In the global retraction $\mathcal{R}$, every higher-dimensional push fixes F. The push on F is exactly this radial projection, and all other pushes fix its three-dimensional boundary. Therefore

$$(\mathcal{R} \circ f)_*[z(R,S)] = [\partial F]. \tag{14}$$

The right-hand classes generate $H_3(Q)$ by (11), and $(\mathcal{R} \circ f)_*$ is an isomorphism by Proposition 2.4. This proves the assertion with one chosen pentagon per face. □

2.3 A five-coloring annihilates the generators

For a finite loopless graph G, the neighborhood complex $\mathcal{N}(G)$ consists of the nonempty vertex families with a common graph neighbor. For G_n, complements are within $[2n]$, and

$$A \sim B \quad \Longleftrightarrow \quad |A \cap B| \leq 1 \quad \Longleftrightarrow \quad d_J(A, B^c) \leq 1.$$

Thus

$$\mathcal{N}(G_n) = \mathcal{C}(2n, n). \tag{15}$$

Suppose $\kappa : V(G_n) \to [5]$ is proper. Every common-neighborhood simplex misses the color of a common neighbor, so κ induces a simplicial map

$$\kappa_{\mathcal{N}} : \mathcal{C}(2n, n) \longrightarrow \mathcal{N}(K_5) = \partial\Delta^4 \simeq S^3. \tag{16}$$

Proposition 2.7. *The map $\kappa_{\mathcal{N}}$ induces the zero homomorphism on third homology over $\mathbb{F}_2$.*

Proof. For a generator of the first type in (12), restrict κ to the ten vertices $R \cup e$, $e \in \binom{S}{2}$, and regard this as an edge-coloring of K_5 on S. For each pair $p \subset S$, all vertices corresponding to edges in $I(p)$ lie in the Johnson ball centered at $R \cup p$. This is a simplex of $\mathcal{C}(2n, n)$, so it uses at most four colors. Lemma 2.2 says that the five vertices of the chosen pentagon are not rainbow.

Their images thus lie in a single simplex of $\partial\Delta^4$ on at most four vertices. The entire image of the boundary cycle is contained in that contractible simplex, so its class in $H_3(\partial\Delta^4)$ is zero.

For the second type, use the ten vertices $R \cup (S \setminus e)$. The edges in $I(p)$ correspond to vertices in the ball centered at $R \cup (S \setminus p)$, giving the same conclusion. Theorem 2.6 now proves that $(\kappa_{\mathcal{N}})_* = 0$ on all third homology. □

2.4 Why the induced map must be nonzero

For a graph G, let $X(G) = \mathrm{Hom}(K_2, G)$ be the complex whose cells are $\Delta(A) \times \Delta(B)$ for nonempty vertex sets A, B such that every vertex of A is adjacent to every vertex of B. The sets A, B are its two shores. Its points are pairs of probability vectors (x, y) on $V(G)$ with completely adjacent supports; the support of a probability vector is the set of its positive coordinates. The shore swap $(x, y) \mapsto (y, x)$ is an involution, meaning its square is the identity. It is free, meaning it has no fixed point, because G has no loops. A map between spaces with involutions is equivariant when it commutes with them. On S^j, the involution is the antipodal map $v \mapsto -v$; a sphere self-map is odd when $f(-v) = -f(v)$. The following is the standard extension and odd-degree argument in this language [22, 23, 2, 18].

Lemma 2.8 (Color-induced nonvanishing)**.** *Let $t \geq 1$, and let G be a finite loopless graph whose neighborhood complex is $(t-1)$-connected. Every proper coloring $\kappa : G \to K_{t+2}$ induces a nonzero homomorphism*

$$(\kappa_{\mathcal{N}})_* : H_t(\mathcal{N}(G); \mathbb{F}_2) \longrightarrow H_t(\mathcal{N}(K_{t+2}); \mathbb{F}_2) \cong \mathbb{F}_2.$$

Proof. The first projection $p_G : X(G) \to |\mathcal{N}(G)|$, $(x, y) \mapsto x$, is a homotopy equivalence [2, Proposition 4.2]. Let u_A be the uniform probability vector on A, with entries $1/|A|$ on A and zero elsewhere. Face posets order nonempty faces by inclusion, and the product cells here are ordered coordinatewise. The barycentric subdivision is the triangulation whose vertices are face barycenters and whose simplexes correspond to chains of nested faces. On these subdivisions the projection is the face-poset map $(A, B) \mapsto A$: the cell barycenter (u_A, u_B) projects to u_A, and both maps are affine on each subdivision simplex. For a face σ of $\mathcal{N}(G)$, choose a common neighbor b. The lower fiber over σ, consisting of cells (A, B) with $A \subseteq \sigma$, contracts by the comparable order-preserving maps

$$(A, B) \leq (A, B \cup \{b\}) \geq (A, \{b\}) \leq (\sigma, \{b\}), \qquad A \subseteq \sigma.$$

Quillen's Theorem A [24], in its finite-poset form [4, Theorem 4.1], gives the claimed equivalence for this particular projection.

Define $X(\kappa)(x, y) = (x', y')$ by $x'_j = \sum_{\kappa(v)=j} x_v$ and $y'_j = \sum_{\kappa(v)=j} y_v$. This pushes the probability coordinates to their colors and gives the following identity, expressing compatibility with the projections

$$p_{K_{t+2}} \circ X(\kappa) = |\kappa_{\mathcal{N}}| \circ p_G.$$

Thus $X(G)$ is $(t-1)$-connected. There is an equivariant map $a : S^t \to X(G)$: start with an antipodal pair and extend successively over one hemisphere of each S^j, for $1 \leq j \leq t$, using

$(t-1)$-connectedness; equivariance defines the opposite hemisphere. Here the extension for $j = 1$ uses path-connectedness, and the later extensions use $\pi_{j-1}(X(G)) = 0$.

Write $\|z\|_1 = \sum_i |z_i|$. The map $(x, y) \mapsto x - y$ identifies $X(K_{t+2})$ equivariantly with

$$\left\{ z \in \mathbb{R}^{t+2} : \sum_i z_i = 0,\ \|z\|_1 = 2 \right\},$$

the boundary of a centrally symmetric convex body in a $(t+1)$-dimensional vector space. Radial normalization identifies this boundary with the antipodal sphere S^t. Consequently $X(\kappa) \circ a$ is an odd self-map of S^t. The degree of a sphere self-map is the integer by which it multiplies the top-dimensional integral homology class. Its degree is odd by Borsuk's theorem [12, Proposition 2B.6], so its map on $H_t(-; \mathbb{F}_2)$ is nonzero. The compatibility identity above and the homology isomorphisms induced by the first projections give the assertion. □

Proof of Theorem 1.1. For $n = 2$, the graph is K_6. The upper bound was established at the start of this section. For $n \geq 3$, suppose a proper five-coloring exists. By (15) and Theorem 2.6, its neighborhood complex is 2-connected. Lemma 2.8 forces the color-induced map on H_3 to be nonzero. Proposition 2.7 makes the same map zero. This contradiction excludes five colors, and the explicit six-coloring proves the theorem. □

A complex L is a simplicial retract of K if there are simplicial maps $j : L \to K$ and $p : K \to L$ whose composition is the identity map on L, written $p \circ j = \mathrm{id}_L$.

Proposition 2.9 (Third homology persists)**.** *For every $n \geq 3$, the complex $\mathcal{C}(6,3)$ is a simplicial retract of $\mathcal{C}(2n,n)$. In particular,*

$$H_3(\mathcal{C}(6,3); \mathbb{F}_2) \cong \mathbb{F}_2^{11} \quad \textit{is a direct summand of} \quad H_3(\mathcal{C}(2n,n); \mathbb{F}_2).$$

Consequently $\mathcal{N}(\mathrm{KG}(2n,n,2))$ is not 3-connected for any $n \geq 3$.

Proof. Choose a six-element set $S \subseteq [2n]$ and a set $R \subseteq [2n] \setminus S$ with $|R| = n-3$. The vertex map

$$j : \binom{S}{3} \longrightarrow \binom{[2n]}{n}, \qquad j(A) = R \cup A,$$

preserves Johnson distance and hence gives a simplicial map $j : \mathcal{C}(6,3) \to \mathcal{C}(2n,n)$. Order S. Balance a subset to size three by adding its least missing elements when it is too small, or deleting its least elements when it is too large. This defines $\rho_3 : 2^S \to \binom{S}{3}$, the map whose nonexpansion is proved in Lemma 4.2. Define

$$p : \binom{[2n]}{n} \longrightarrow \binom{S}{3}, \qquad p(B) = \rho_3(B \cap S).$$

Restriction to S is non-expanding for $D(X,Y) = \max\{|X \setminus Y|, |Y \setminus X|\}$, as is ρ_3. On an equal-size layer, D is Johnson distance. Therefore, if a face σ of $\mathcal{C}(2n,n)$ has center Z, then

$$d_J(p(B), p(Z)) \leq D(B \cap S, Z \cap S) \leq d_J(B, Z) \leq 1 \qquad (B \in \sigma).$$

Thus p is simplicial. Since balancing fixes every three-element set, $p \circ j = \mathrm{id}$.

Apply the polyhedral model to $P = \Delta(6,3)$, with Q the retained subcomplex of its three-skeleton and simplex faces. The relative cellular complex has twelve four-cells, the facets $x_i = 0$ and $x_i = 1$, and one five-cell, the interior of P. There are no relative three-cells. Over $\mathbb{F}_2$, the boundary of the five-cell is the sum of the twelve distinct four-cells and is nonzero. Hence

$$H_4(P, Q; \mathbb{F}_2) \cong \mathbb{F}_2^{12} / \langle (1, \ldots, 1) \rangle \cong \mathbb{F}_2^{11}.$$

The affine-map comparison and radial retraction identify this group with $H_3(\mathcal{C}(6,3);\mathbb{F}_2)$. The simplicial retraction makes j_* split injective, so the displayed group is a direct summand for every $n \geq 3$. Since a 3-connected space has zero third homology, the final assertion follows from $\mathcal{N}(\mathrm{KG}(2n,n,2)) = \mathcal{C}(2n,n)$. □

Equivariant index. Let $\mathbb{Z}_2$ be the two-element group; a free $\mathbb{Z}_2$-space is a space equipped with a free involution. For such a space X, write $\mathrm{ind}_{\mathbb{Z}_2}(X)$ for the least d admitting an equivariant map $X \to S^d$, where S^d has the antipodal action. For $n \geq 3$, the present arguments give

$$3 \leq \mathrm{ind}_{\mathbb{Z}_2} \mathrm{Hom}(K_2, \mathrm{KG}(2n,n,2)) \leq 4.$$

The lower bound follows from the equivariant map $S^3 \to \mathrm{Hom}(K_2, G_n)$; the upper bound follows from the six-coloring and the equivariant homeomorphism from $\mathrm{Hom}(K_2, K_6)$ to the antipodal sphere S^4.

Question 2.10. For $n \geq 3$, is $\mathrm{ind}_{\mathbb{Z}_2} \mathrm{Hom}(K_2, G_n)$ equal to three or four?

3 Monotone valuations and sharp balanced thresholds

Let M be a finite set of indivisible goods. There are two fixed, nonempty groups of agents, G_1 and G_2. An allocation is a subset $A \subseteq M$: the first group receives A, and the second receives $A^c = M \setminus A$. Every agent compares the two group bundles using a valuation $u : 2^M \to \mathbb{R}_{\geq 0}$. We assume normalization, meaning $u(\varnothing) = 0$, and monotonicity: $u(S) \leq u(T)$ when $S \subseteq T$. An additive valuation has the form $u(S) = \sum_{g \in S} w(g)$ with $w(g) \geq 0$.

Definition 3.1. An agent who receives P and compares it with Q is *envy-free up to c goods*, or EFc, if some $T \subseteq Q$ with $|T| \leq c$ satisfies

$$u(P) \geq u(Q \setminus T).$$

An allocation is EFc if it is EFc for every agent. It is *balanced* when the two bundles have the same size, and *almost balanced* when their sizes differ by at most one.

An agent rejects an allocation when it fails EFc for that agent. An independent set of vertices contains no adjacent pair. The graph method rests on the observation that an agent cannot reject both ends of an edge of the graph below.

Lemma 3.2 (Independent rejection families)**.** *For a fixed monotone agent, the allocations at which that agent fails EFc form an independent set in the graph on 2^M with adjacency*

$$|A \cap B| \leq c, \qquad |A^c \cap B^c| \leq c.$$

When $|M| = 2r$ and $r \geq c+1$, its balanced layer is $\mathrm{KG}(2r, r, c+1)$.

Proof. Suppose a first-group agent rejects adjacent allocations A and B. Deleting $A^c \cap B^c$ from the other bundle at A gives

$$u(A) < u(A^c \cap B) \leq u(B).$$

Reversing A and B gives $u(B) < u(A)$, a contradiction. For a second-group agent, exchange the bundles with their complements. Finally, for two r-sets in a $2r$-element set, $|A^c \cap B^c| = |A \cap B|$. □

If every allocation in a chosen configuration graph is rejected, assigning each vertex to an agent who rejects it gives a proper coloring. Thus fewer agents than colors force an EFc allocation. For the balanced graph the converse holds as well. We recall the threshold principle of Kyropoulou–Suksompong–Voudouris [19, Theorem 4.4 and the discussion after Corollary 4.5], and give the construction needed later.

Theorem 3.3 (Balanced EFc threshold [19])**.** *Fix $c \geq 1$ and $r \geq c+1$. Let $T^{\mathrm{bal}}_{r,c}$ be the largest integer t such that every monotone instance on $2r$ goods, with two fixed groups and at most t agents in total, has a balanced EFc allocation. Then*

$$T^{\mathrm{bal}}_{r,c} = \chi(\mathrm{KG}(2r, r, c+1)) - 1.$$

If $y = \chi(\mathrm{KG}(2r, r, c+1))$, every prescribed split of y agents admits monotone $0/1$ valuations for which no balanced EFc allocation exists.

Proof. Put $K = \mathrm{KG}(2r, r, c+1)$, $y = \chi(K)$, and $M = [2r]$. We first prove the existence guarantee. Consider any two-group instance with at most $y-1$ agents. Represent a balanced allocation by an r-set $X \subseteq M$, with the first group receiving X and the second group receiving X^{c}. For each agent a, let $B_a \subseteq \binom{M}{r}$ be the family of balanced allocations at which a is not EFc.

Each B_a is independent in K by Lemma 3.2. Hence fewer than y rejection families cannot cover $\binom{M}{r}$; otherwise, assigning each balanced allocation to one agent who rejects it would give a proper coloring of K with fewer than y colors. Thus every such instance admits a balanced EFc allocation, and

$$T^{\mathrm{bal}}_{r,c} \geq y - 1.$$

For sharpness, fix a proper coloring

$$\varphi : \binom{M}{r} \longrightarrow [y]$$

that uses all y colors, and fix any prescribed split of y agents between the two groups. Assign the colors bijectively to the agents. For a color j assigned to a first-group agent, define the monotone $0/1$ valuation

$$u_j(S) = 0 \quad \Longleftrightarrow \quad S \subseteq A \text{ for some } A \in \binom{M}{r} \text{ with } \varphi(A) = j.$$

For a color j assigned to a second-group agent, define

$$u_j(S) = 0 \quad \Longleftrightarrow \quad S \subseteq A^{\mathrm{c}} \text{ for some } A \in \binom{M}{r} \text{ with } \varphi(A) = j.$$

In both cases the zero sets are downward closed, so the valuation is monotone.

Consider a balanced allocation represented by A, and put $j = \varphi(A)$. If color j belongs to a first-group agent, then $u_j(A) = 0$. For every $T \subseteq A^{\mathrm{c}}$ with $|T| \leq c$, we have $u_j(A^{\mathrm{c}} \setminus T) = 1$: otherwise $A^{\mathrm{c}} \setminus T \subseteq B$ for some B of color j, which would imply $|A \cap B| \leq |T| \leq c$, contradicting properness of φ. Hence that agent is not EFc at A.

If color j belongs to a second-group agent, then $u_j(A^{\mathrm{c}}) = 0$. For every $T \subseteq A$ with $|T| \leq c$, we have $u_j(A \setminus T) = 1$: otherwise $A \setminus T \subseteq B^{\mathrm{c}}$ for some B of color j, so $A \cap B \subseteq T$ and again $|A \cap B| \leq c$, a contradiction. Thus this agent is not EFc at A.

Every balanced allocation is therefore rejected by the agent carrying its color. This gives a counterexample for the prescribed split with y agents, so $T^{\mathrm{bal}}_{r,c} \leq y - 1$. Combining the two inequalities proves the theorem. □

Corollary 3.4 (Sharp balanced EF1)**.** *For every $r \geq 2$, every monotone instance on $2r$ goods with at most five agents has a balanced EF1 allocation. For every prescribed split of six agents, there is such an instance with no balanced EF1 allocation.*

Proof. Combine Theorems 1.1 and 3.3. □

Proposition 3.5 (Explicit six-agent counterexamples)**.** *For every $r \geq 2$ and every prescribed split of six agents, there are normalized monotone $0/1$ valuations on $2r$ goods with no balanced EF1 allocation.*

Proof. Let M be a set of $2r$ goods. Fix three distinct goods in M and let e range over their three pairs. For a statement E, the scalar indicator $\mathbf{1}_{\{E\}}$ is one when E holds and zero otherwise. For $S \subseteq M$, put

$$f_e(S) = \mathbf{1}_{\{|S \cup e| > r\}}, \qquad g_e(S) = \mathbf{1}_{\{S \cap e \neq \varnothing\}}.$$

Both are normalized monotone $0/1$ valuations. The six families $\{A \in \binom{M}{r} : e \subseteq A\}$ and $\{A \in \binom{M}{r} : e \subseteq A^c\}$ cover the balanced layer. Assign one family to each agent in any prescribed split, and use the following valuation for that agent:

Assigned family	First-group agent	Second-group agent
$e \subseteq A$	f_e	g_e
$e \subseteq A^c$	g_e	f_e

At every allocation in the assigned family, the own bundle has value zero. For g_e, the other bundle contains both goods of e, so one survives any single deletion. For f_e, that bundle avoids e and retains at least $r - 1$ goods, so its union with e has size at least $r + 1$. Thus the other bundle retains value one after any allowed deletion. Every balanced allocation is rejected. This covers $(1, 5)$, $(2, 4)$, $(3, 3)$ and their reversals. □

Corollary 3.6 (Odd numbers of goods)**.** *Every monotone instance with at most five agents and any number of goods has an almost-balanced EF1 allocation.*

Proof. For at most two goods, give each group at most one good. Otherwise, if the number of goods is odd, add a dummy good $*$ and extend each valuation by $u'(S) = u(S \setminus \{*\})$. Apply Corollary 3.4 and remove the dummy. The two sizes differ by one. Any deletion witnessing EF1 before removal still witnesses EF1 after its dummy element, if present, is removed. The even case is Corollary 3.4. □

Corollary 3.7 (Allocator preferences)**.** *Given four monotone valuations v_1, u_1, v_2, u_2 on an even number of goods, there is a balanced allocation satisfying EF1 for both v_1, u_1 on the first bundle and both v_2, u_2 on the second. More generally, a fifth prescribed monotone comparison may be imposed on either bundle. For an odd number of goods, the allocation can be almost balanced.*

Proof. Regard each comparison as an agent in its prescribed group and apply Corollaries 3.4 and 3.6. □

The four-comparison case is a balanced version of the two-agent guarantee requiring EF1 simultaneously for agent utilities and allocator utilities in Bu–Li–Liu–Song–Tao's allocator-preference model [6]. Writing the comparisons explicitly also permits the fifth constraint without introducing a separate allocation model.

For larger c, Daneshpajouh's lower bound [7] gives

$$\chi(\mathrm{KG}(2r, r, c + 1)) \geq 2c + 2,$$

so the same principle guarantees balanced EFc for at most $2c+1$ agents. When $r \le c$, every balanced allocation is EFc, since the entire comparison bundle may be deleted. The EF1 theorem improves the general $2c+1$ guarantee from three agents to five.

4 Balancing the full allocation graph

For $m \ge 2c+1$, with complements taken within $[m]$, let $\Gamma_{c,c}(m)$ be the loopless graph on $2^{[m]}$ whose edges satisfy

$$|A \cap B| \le c, \qquad |A^c \cap B^c| \le c.$$

By Lemma 3.2, each monotone agent's rejection family is independent in this graph. For even m, its balanced layer has the same chromatic number as the whole graph.

Theorem 4.1 (Allocation–Kneser equivalence)**.** *For $c \ge 0$ and $m \ge 2c+1$,*

$$\chi(\Gamma_{c,c}(m)) = \chi\left(\mathrm{KG}\left(2\left\lceil \frac{m}{2}\right\rceil, \left\lceil \frac{m}{2}\right\rceil, c+1\right)\right).$$

When $m = 2r$, the balanced layer includes into the allocation graph, and an explicit homomorphism in the other direction fixes that layer pointwise.

Let $r \ge 1$ be an integer. For a set U of size $2r$, fix once and for all a linear order on U. Define a balancing map

$$\rho : 2^U \longrightarrow \binom{U}{r}$$

as follows. If $|X| \le r$, let $\rho(X)$ be obtained from X by adding the smallest $r - |X|$ elements of $U \setminus X$. If $|X| > r$, let $\rho(X)$ be obtained from X by deleting the smallest $|X| - r$ elements of X. Then

$$|\rho(X)| = r \qquad \text{and} \qquad \rho(U \setminus X) = U \setminus \rho(X).$$

Lemma 4.2 (Balancing is non-expanding)**.** *For all $X, Y \subseteq U$,*

$$\max\{|\rho(X) \setminus \rho(Y)|, |\rho(Y) \setminus \rho(X)|\} \le \max\{|X \setminus Y|, |Y \setminus X|\}.$$

Proof. Realize ρ by scanning the elements of U in the fixed order. At a scanned element e, a current set Z is changed only if it is still too small and misses e, in which case e is added, or if it is still too large and contains e, in which case e is deleted.

It is enough to check that one scan step is non-expanding for

$$D(X,Y) := \max\{|X \setminus Y|, |Y \setminus X|\}.$$

Only the scanned coordinate can change. Suppose, after the step, $|X \setminus Y|$ increases by one. Then the step has created a new element of $X \setminus Y$, so before the step the set Y was larger than X. Hence

$$|Y \setminus X| - |X \setminus Y| = |Y| - |X| \ge 1,$$

and the increased value $|X \setminus Y| + 1$ is still at most the old value of $|Y \setminus X|$, hence at most the old maximum $D(X,Y)$. The same argument with X and Y interchanged handles a possible increase of $|Y \setminus X|$. Thus every scan step is non-expanding, and so is their composition ρ. □

Proposition 4.3 (Even balancing homomorphism)**.** *For $r \ge c+1$, the balancing map gives a graph homomorphism*

$$\Gamma_{c,c}(2r) \longrightarrow \mathrm{KG}(2r, r, c+1).$$

Proof. Let $U = [2r]$. Suppose $A \sim B$ in $\Gamma_{c,c}(2r)$. Equivalently,

$$|A \cap B| \le c \qquad \text{and} \qquad |A^{\mathrm{c}} \cap B^{\mathrm{c}}| \le c.$$

Put $R = B^{\mathrm{c}}$. Then

$$|A \setminus R| = |A \cap B| \le c, \qquad |R \setminus A| = |B^{\mathrm{c}} \cap A^{\mathrm{c}}| \le c.$$

By Lemma 4.2,

$$|\rho(A) \setminus \rho(R)| \le c.$$

Using $\rho(R) = \rho(B^{\mathrm{c}}) = \rho(B)^{\mathrm{c}}$, we obtain

$$|\rho(A) \cap \rho(B)| = |\rho(A) \setminus \rho(B)^{\mathrm{c}}| = |\rho(A) \setminus \rho(R)| \le c.$$

Thus $\rho(A)$ and $\rho(B)$ are adjacent in $\mathrm{KG}(2r, r, c+1)$. □

Lemma 4.4 (Odd extension)**.** *For $r \ge c+1$, choose a new element $\infty \notin [2r-1]$. The map*

$$\theta : 2^{[2r-1]} \longrightarrow 2^{[2r-1]\cup\{\infty\}}, \qquad \theta(A) = \begin{cases} A \cup \{\infty\}, & |A| \le r-1, \\ A, & |A| \ge r, \end{cases}$$

is a homomorphism

$$\Gamma_{c,c}(2r-1) \longrightarrow \Gamma_{c,c}(2r).$$

Proof. Let $A \sim B$ in $\Gamma_{c,c}(2r-1)$, and put

$$p = |A \cap B|, \qquad q = |A^{\mathrm{c}} \cap B^{\mathrm{c}}|.$$

Then $p, q \le c$ and

$$|A| + |B| = (2r-1) + p - q.$$

The new point ∞ can create one additional common chosen coordinate only when $|A|, |B| \le r-1$. If already $p = c$, then

$$|A| + |B| = (2r-1) + c - q \ge 2r-1,$$

contradicting $|A| + |B| \le 2r-2$. Hence, whenever ∞ is added to both sets, we have $p \le c-1$, and the new common chosen count is still at most c.

Similarly, ∞ can create one additional common unchosen coordinate only when $|A|, |B| \ge r$. If already $q = c$, then

$$|A| + |B| = (2r-1) + p - c \le 2r-1,$$

contradicting $|A| + |B| \ge 2r$. Hence the new common unchosen count is also at most c. Thus $\theta(A) \sim \theta(B)$ in $\Gamma_{c,c}(2r)$. □

Proof of Theorem 4.1. First suppose $m = 2r$. The middle layer $\binom{[2r]}{r}$ of $\Gamma_{c,c}(2r)$ induces $\mathrm{KG}(2r, r, c+1)$: for $|A| = |B| = r$,

$$|A \cup B| = 2r - |A \cap B|,$$

so the two $\Gamma_{c,c}$ adjacency conditions are equivalent to $|A \cap B| \le c$. Hence

$$\chi(\Gamma_{c,c}(2r)) \ge \chi(\mathrm{KG}(2r, r, c+1)).$$

The reverse inequality is Proposition 4.3.

Now suppose $m = 2r - 1$. The two central layers

$$\binom{[2r-1]}{r-1} \cup \binom{[2r-1]}{r}$$

induce a copy of $\mathrm{KG}(2r, r, c+1)$: identify an $(r-1)$-set A with $A \cup \{\infty\}$, and leave an r-set unchanged. A direct check in the three cases (r, r), $(r-1, r)$, and $(r-1, r-1)$ shows that adjacency is exactly image intersection at most c. Thus

$$\chi(\Gamma_{c,c}(2r-1)) \geq \chi(\mathrm{KG}(2r, r, c+1)).$$

For the reverse inequality, compose the odd extension $\Gamma_{c,c}(2r-1) \to \Gamma_{c,c}(2r)$ from Lemma 4.4 with the even balancing homomorphism from Proposition 4.3. □

Corollary 4.5. *For every $m \geq 3$, $\chi(\Gamma_{1,1}(m)) = 6$.*

Proof. Apply Theorems 4.1 and 1.1. □

Corollary 4.6 (Balance costs nothing for the chromatic method)**.** *For $2r$ goods and $r \geq c+1$, the independent-rejection argument guarantees EFc for up to $\chi(\mathrm{KG}(2r, r, c+1)) - 1$ agents both on the balanced layer and on the full allocation graph.*

Proof. Combine Theorem 4.1 with Lemma 3.2. □

This corollary concerns the guarantee supplied by the chromatic method. A color class in the full allocation graph need not be realizable as the rejection family of an agent on either prescribed side. Thus the balanced sharpness theorem does not by itself prove sharpness in the unbalanced model. For EF1, a direct example settles the universal threshold.

Theorem 4.7 (Sharp universal EF1 threshold)**.** *For every $m \geq 3$, five is the largest total number of agents for which all monotone two-group instances on m goods are guaranteed to have an EF1 allocation, without a balance requirement. Failure with six agents is possible already for additive valuations and group sizes $(3, 3)$.*

The existence assertion follows from the six-color theorem; the additive six-agent obstruction is due to [19, Corollary 3.6].

Proof. Existence follows from Corollary 3.6; alternatively, it follows from Corollary 4.5 and Lemma 3.2. For failure, use the three-goods construction of [19, Corollary 3.6]. Choose three goods a, b, c. In each group put three agents, one for each pair $\{a, b\}$, $\{a, c\}$, $\{b, c\}$. The agent indexed by a pair values its two goods at one each and every other good at zero. Every allocation puts some pair into one bundle. The agent for that pair in the opposite group receives value zero, whereas deleting any one good from the comparison bundle leaves value at least one. Thus every allocation fails EF1. Extra goods have value zero for all six agents. □

For $m \leq 2$, both bundles may have size at most one, so EF1 holds for any number of agents. The preceding theorem concerns a universal guarantee over all group splits. Its counterexample uses $(3, 3)$; the every-split sharpness assertion in Corollary 3.4 concerns balanced allocations.

5 Equivalent total-search problems

The search complexity question for classical Kneser graphs was posed in [9]. For $N \geq 2k$, classical Kneser search takes an $(N - 2k + 1)$-coloring of the k-subsets of $[N]$ and asks for disjoint sets with the same color. This problem belongs to the parity-based total-search class PPA; its hardness remains open. The stable-set restriction, in which no two chosen elements are consecutive in a cyclic order of the ground set, gives the Schrijver search problem; it is PPA-complete [13]. Related work connects Kneser search under an extended access model to finding a small agreeable set, a bundle every agent values at least as highly as its complement [14, 21]. We establish the balanced generalized-Kneser analogue for EFc search. To make both problems total on arbitrary circuit inputs, we allow a fair-division solver to return an explicit violation of monotonicity, as in Haviv's formal treatment of resource-allocation search [14].

A search problem specifies allowed outputs for each input. It is total when every input has an allowed output. A promise is an extra restriction on admissible inputs. A polynomial-time many-one search reduction constructs one target instance and converts every valid target solution to a valid source solution, with both transformations taking polynomial time.

Fix $c \geq 1$ and positive group sizes q_1, q_2, and put $q = q_1 + q_2$. Index agents by $[q]$, assigning the first q_1 to group one and the rest to group two. The number of goods is $2r$, where $r \geq c + 1$. A subset of $[2r]$ is encoded by its $2r$-bit characteristic vector. A balanced allocation is therefore an encoding of Hamming weight r, meaning that it has r one-bits.

Definition 5.1 (Balanced Kneser Search)**.** The problem $\mathrm{BKS}_{c,q}(r)$ takes a Boolean circuit C with $2r$ input bits and a fixed number of output bits. Interpret its output as a nonnegative integer z. On an input encoding an r-set, define

$$\varphi(X) = \begin{cases} z, & 1 \leq z \leq q, \\ 1, & \text{otherwise.} \end{cases}$$

On inputs of weight different from r, also set the color to one. The required output is two r-sets X, Y satisfying

$$\varphi(X) = \varphi(Y), \qquad |X \cap Y| \leq c.$$

The inequality $r > c$ guarantees that $X \neq Y$.

This decoding makes every circuit define a genuine q-coloring on the middle layer; neither its output range nor its behavior on other layers is a promise. The problem is total whenever $q < \chi(\mathrm{KG}(2r, r, c + 1))$.

For valuation circuits, we use an equally explicit exact-number encoding. Each circuit has a numerator sign bit s, an unsigned numerator-magnitude field a, and an unsigned denominator field b. Decode the output as $(-1)^s a/(b + 1)$. This sign-and-magnitude convention gives every bit string a numerical meaning. Thus every circuit output denotes a finite signed rational number. Integer-valued circuits are a special case. Comparison is exact by cross multiplication with positive denominators. Let L denote the total circuit description length, including the widths of all output fields.

Definition 5.2 (Totalized balanced EFc search)**.** An input consists of one such valuation circuit $u_i : 2^{[2r]} \to \mathbb{Q}$ for every agent in the prescribed split (q_1, q_2). No monotonicity, normalization, or nonnegativity promise is imposed. A valid output is either:

(i) an r-set X, together with, for every agent i, a deletion set D_i of at most c goods from the other group's bundle, witnessing that the allocation (X, X^c) is EFc for that agent; or

(ii) an agent i and sets $S \subsetneq T \subseteq [2r]$ with $u_i(S) > u_i(T)$.

Both output types have polynomial length and can be checked in polynomial time. The second is a concrete certificate that an input valuation is not monotone. If all valuations are monotone, subtracting $u_i(\varnothing)$ normalizes them and makes them nonnegative without changing any EFc comparison. The balanced threshold theorem then provides the first type of output in the chromatic total range. Thus this version of fair-division search is total on arbitrary valuation circuits in that range.

Theorem 5.3 (Search equivalence)**.** *Fix c, q_1, q_2 and suppose $r \geq c+1$ and*

$$q < \chi(\mathrm{KG}(2r, r, c+1)), \qquad q = q_1 + q_2.$$

Balanced Kneser Search $\mathrm{BKS}_{c,q}(r)$ *is equivalent to totalized balanced EFc search for the prescribed split under deterministic polynomial-time many-one reductions. The reductions take time $L^{O(1)}r^{O(c)}$ and produce circuits of size $L^{O(1)}r^{O(c)}$.*

Proof. Put $M = [2r]$. First start with arbitrary valuation circuits. For every balanced allocation X, test all deletion sets of size at most c in each comparison bundle. If some agent rejects the allocation, color X by the least such agent's index; if no agent rejects it, use color one. For fixed c this is a circuit of size $L^{O(1)}r^{O(c)}$.

Let X, Y be the monochromatic edge returned by Balanced Kneser Search. Test both allocations. If either is fair, output it and the deletion sets found during these tests. Otherwise both allocations are rejected by the agent i carrying their common color, including when that color is one. Let P_X, P_Y be this agent's own bundles at the two allocations, and $Q_X = M \setminus P_X$, $Q_Y = M \setminus P_Y$ the comparison bundles. Both pairs are balanced, and

$$|Q_X \cap Q_Y| = |X \cap Y| \leq c.$$

Badness gives

$$u_i(P_X) < u_i(S_X), \qquad u_i(P_Y) < u_i(S_Y), \quad S_X = Q_X \cap P_Y, \quad S_Y = Q_Y \cap P_X,$$

by deleting $Q_X \cap Q_Y$ from each comparison bundle. Since $S_X \subseteq P_Y$ and $S_Y \subseteq P_X$, at least one of

$$u_i(S_X) > u_i(P_Y), \qquad u_i(S_Y) > u_i(P_X)$$

must hold. Indeed, if neither held, the preceding strict inequalities would yield

$$u_i(P_X) < u_i(S_X) \leq u_i(P_Y) < u_i(S_Y) \leq u_i(P_X),$$

a contradiction. Return whichever nested pair violates monotonicity. The containment is proper because a strict inequality is impossible when the two sets are equal. This decodes every target solution into a valid fair-division or monotonicity output.

Conversely, let $\varphi : \binom{M}{r} \to [q]$ be the coloring obtained by the stated circuit decoder. Assign its colors bijectively to the agents. Let I_1 and I_2 be the colors assigned to the first and second groups. For a first-group color $i \in I_1$, define a binary valuation by

$$u_i(S) = 0 \iff |S| < r - c \quad \text{or} \quad \big(|S| \leq r \text{ and } \exists Y \in \tbinom{M}{r} : S \subseteq Y,\ \varphi(Y) = i\big).$$

For a second-group color $i \in I_2$, use

$$u_i(S) = 0 \iff |S| < r - c \quad \text{or} \quad \big(|S| \leq r \text{ and } \exists Y \in \tbinom{M}{r} : Y \subseteq M \setminus S,\ \varphi(Y) = i\big).$$

In both definitions set $u_i(S) = 1$ otherwise. Their zero sets are downward closed, so these are normalized nonnegative monotone valuations. The only nontrivial range is $r - c \leq |S| \leq r$.

There are $r^{O(c)}$ relevant completions of S to an r-set, or r-sets contained in $M \setminus S$, so polynomial-size circuits compute them. The small-bundle clause is what avoids enumerating exponentially many completions for smaller S.

Any valid output of totalized EFc search for these circuits must be a fair allocation, since no monotonicity violation exists. Let X be its first-group bundle and put $i = \varphi(X)$. If $i \in I_1$, then $u_i(X) = 0$. Use the returned deletion witness $D_i \subseteq X^c$, with $|D_i| \leq c$. Fairness forces $u_i(X^c \setminus D_i) = 0$. This set has size at least $r - c$, so the small-bundle clause does not apply. Hence some color-i r-set Y contains $X^c \setminus D_i$. Its intersection with X has size at most $|D_i| \leq c$.

If $i \in I_2$, then $u_i(X^c) = 0$. The returned witness $D_i \subseteq X$ gives $u_i(X \setminus D_i) = 0$. Again its size is at least $r - c$, so some color-i r-set satisfies

$$Y \subseteq M \setminus (X \setminus D_i) = X^c \cup D_i.$$

Thus $|X \cap Y| \leq c$. In either case, enumerate the $r^{O(c)}$ candidate completions to recover Y. The pair X, Y is a monochromatic edge, and decoding is polynomial-time for fixed c. □

Corollary 5.4 (Five-color search)**.** *For every fixed positive split (q_1, q_2) with $q_1 + q_2 = 5$, totalized balanced EF1 search is polynomial-time equivalent to finding a monochromatic edge in a circuit-specified five-coloring of* $\mathrm{KG}(2r, r, 2)$*, for every $r \geq 2$.*

Proof. Apply Theorems 1.1 and 5.3. □

In particular, these five-color problems are syntactically total search problems with polynomially verifiable outputs. For general c, $\chi(\mathrm{KG}(2r, r, c+1)) \geq 2c + 2$ gives the same uniform totality and equivalence with $2c + 1$ colors and agents. On monotone inputs the totalized problem must return a fair allocation, so the reductions also recover the original equivalence under promised monotone valuation access. The reductions take $L^{O(1)}r^{O(c)}$ time, which is polynomial for fixed c. When c varies, their stated bounds have an exponent depending on c, the dependence called XP in parameterized complexity. A fixed-parameter bound would use a function f depending only on c and a constant exponent a independent of c, and would take the form $f(c)(L + r)^a$; no such bound is established here. The equivalence transfers algorithms or complexity classifications between the two search tasks. It supplies neither a polynomial-time algorithm for solving them nor a proof of PPA membership or hardness.

The colorability-defect route. A standard topological route to Kneser-search membership uses the colorability defect of its defining set family [15]. For a family $\mathcal{H}$ of nonempty subsets of a finite ground set U, the two-colorability defect $\mathrm{cd}_2(\mathcal{H})$ is the minimum number of elements deleted from U so that the remaining elements can be colored red and blue without making any surviving member of $\mathcal{H}$ monochromatic. A member survives when it is contained entirely in the remaining ground set. This is a coloring of ground-set elements, distinct from proper coloring of the corresponding disjointness graph. The disjointness graph of a set family has its members as vertices and joins the disjoint pairs. The natural set-family representation here has zero defect for $r \geq 4$.

Proposition 5.5. *Put $U = \binom{[2r]}{2}$ and $\mathcal{H}_r = \{\binom{A}{2} : A \in \binom{[2r]}{r}\}$. The disjointness graph on $\mathcal{H}_r$ is* $\mathrm{KG}(2r, r, 2)$*, but $\mathrm{cd}_2(\mathcal{H}_r) = 0$ for every $r \geq 4$.*

Proof. Two members $\binom{A}{2}, \binom{B}{2}$ are disjoint exactly when $|A \cap B| \leq 1$. To see that the family is already two-colorable, partition $[2r]$ into parts of sizes $r - 1, r - 1, 2$. Color a pair red within a part and blue across parts. A red clique has size at most $r - 1$, and a blue clique has size at most three. No r-set has all its pairs of one color. Thus no ground-set elements need be deleted to obtain a two-coloring, which is the definition of zero two-colorability defect. □

This representation therefore supplies no colorability-defect certificate for the five-color search problem. This is a limitation of the stated encoding, rather than an obstruction to PPA membership by another route. A PPA reduction would need a further argument beyond this encoding.

Question 5.6. Does the five-color Balanced Kneser Search problem belong to PPA? A membership result, or a hardness result, would transfer to its totalized EF1 formulation by the reductions above.

6 One additive agent and six-agent EF2

An additive valuation allows us to make one agent fair throughout a smaller family of balanced allocations. The remaining obstruction lies in a Hamming graph. For binary words $x, y \in \{0,1\}^r$, define their Hamming distance by $d_H(x,y) = |\{j \in [r] : x_j \neq y_j\}|$. For $1 \leq \delta \leq r$, write $\mathrm{H}_2(r,\delta)$ for the graph on $\{0,1\}^r$ whose distinct vertices are adjacent at Hamming distance at least δ.

Lemma 6.1 (Trace to the Hamming cube)**.** *For every $r \geq 3$,*

$$\chi(\mathrm{H}_2(r, r-2)) \geq 6.$$

Proof. For $r = 3$, the graph is K_8. For $r \geq 4$, put $n = \lceil r/2 \rceil$ and fix $S \subseteq [2n]$ of size r. Map an n-set A to the characteristic vector of $A \cap S$. If A and B are adjacent in $\mathrm{KG}(2n, n, 2)$, then

$$\begin{aligned} d_H(\mathbf{1}_{A\cap S}, \mathbf{1}_{B\cap S}) &= |(A \triangle B) \cap S| \\ &\geq |A \triangle B| - (2n - r) \\ &\geq (2n-2) - (2n-r) = r - 2. \end{aligned}$$

The images are distinct, and this map is a graph homomorphism. Apply Theorem 1.1. □

Theorem 6.2 (One-additive-agent transfer)**.** *Let $r > c \geq 1$. Suppose a two-group instance on $2r$ goods has t agents, one with an additive valuation and all others with monotone valuations. If*

$$t \leq \chi(\mathrm{H}_2(r, r-c)),$$

then it has a balanced EFc allocation.

Proof. Choose the additive agent a_0 and sort the goods by its weights:

$$w(g_1) \geq w(g_2) \geq \cdots \geq w(g_{2r}).$$

Pair consecutive goods, $P_j = \{g_{2j-1}, g_{2j}\}$. Call allocations that give each group one good from every pair transversal allocations. We consider only these allocations. They are balanced. Every such allocation is EF1 for a_0. Indeed, delete the other bundle's good in the first pair. Its good from pair j, for $j \geq 2$, has weight at most that of the own bundle's good from pair $j-1$. These comparisons match all remaining goods in the other bundle to distinct goods in the own bundle. Nonnegative weights prove EF1, and hence EFc.

Encode a transversal allocation by a binary word recording the first group's choice from each pair, and use the same letters for allocations and their encoded words. For two transversals A, B,

$$|A \cap B| = r - d_H(A, B).$$

Their conflict graph is therefore $\mathrm{H}_2(r, r-c)$. By Lemma 3.2, each of the other $t-1$ agents rejects an independent family in this graph. Since $t - 1 < \chi(\mathrm{H}_2(r, r-c))$, these families cannot cover all transversals. One transversal is fair for them all, and it is already fair for a_0. □

This argument uses the consecutive-pair idea from the additive group allocation work of Gölz and Yaghoubizade [11]. Only the selected agent needs additivity.

Corollary 6.3 (Six agents, one additive valuation)**.** *Every two-group instance with at most six agents has a balanced EF2 allocation when the number of goods is even, provided at least one valuation is additive and the others are monotone. For an odd number of goods, it has an almost-balanced EF2 allocation.*

Proof. With $2r$ goods, the assertion is immediate for $r \leq 2$. For $r \geq 3$, use Lemma 6.1 and Theorem 6.2. For odd numbers, add a zero-weight dummy good as in Corollary 3.6; the selected valuation remains additive. □

The projection in Lemma 6.1 makes the six-agent EF2 guarantee another consequence of the same six-color theorem.

7 Explicit obstructions for higher EFc

Balogh, Cherkashin, and Kiselev used discrepancy to color generalized Kneser graphs [3]. Hadamard constructions were also studied by prior work [16]. We use the signed-witness form of that method to obtain small explicit colorings, then turn them into fair-division counterexamples.

Lemma 7.1 (Transfer from a finite balanced layer)**.** *For $n \geq r \geq d \geq 1$,*

$$\mathrm{KG}(2n, n, d) \longrightarrow \mathrm{KG}(2r, r, d).$$

Proof. Restrict each n-set to a fixed $2r$-element ground set. On an edge, both the common chosen part and the common unchosen part have size at most $d-1$, and restriction preserves these bounds. Thus the trace maps into $\Gamma_{d-1,d-1}(2r)$. Composing with the balancing homomorphism of Proposition 4.3 proves the claim. □

For a $2n$-element ground set, an n-subset is called a half-set. For $A, F \subseteq [2n]$ with A a half-set, the discrepancy on F is the imbalance $|A \cap F| - |A^c \cap F| = 2|A \cap F| - |F|$.

Lemma 7.2 (Signed discrepancy witnesses [3])**.** *Fix integers $1 \leq d \leq n$, and let $\mathcal{F} \subseteq 2^{[2n]}$. Suppose that every n-set A has a witness $F \in \mathcal{F}$ satisfying*

$$|A \cap F| \geq \frac{|F| + d}{2} \quad \text{or} \quad |A^c \cap F| \geq \frac{|F| + d}{2}.$$

A witness is positive when the first inequality holds and negative when the second holds. Then $\chi(\mathrm{KG}(2n, n, d)) \leq 2|\mathcal{F}|$.

Proof. Color A by an available signed witness. Two sets with the same positive witness satisfy

$$|A \cap B| \geq |A \cap B \cap F| \geq |A \cap F| + |B \cap F| - |F| \geq d.$$

For a negative witness their complements satisfy the same inequality, and balancedness gives $|A \cap B| = |A^c \cap B^c|$. □

For $d = 2$, the three pairs of a triangle give the six-coloring in Section 2. For general d, taking all d-subsets of a fixed $(2d-1)$-set gives the elementary bound

$$\chi(\mathrm{KG}(2n, n, d)) \leq 2\binom{2d-1}{d} \qquad (n \geq d),$$

since one of A, A^c contains at least d of those points. The finite geometries below replace this complete family by smaller witness systems. The signed-witness method and the Fano-plane example are recorded in [3]. The Walsh construction below uses orthogonality of functions taking the values 1 and -1 on a vector space over the two-element field; the projective construction uses incidence counts between points and hyperplanes. Their role here is to give explicit fair-division counterexamples in the stated ranges. The longer proofs appear in Appendix A. The comparison uses the elementary complete-family benchmark.

Proposition 7.3 (Fano-plane coloring [3])**.** *For $n \geq 4$, $\chi(\mathrm{KG}(2n, n, 3)) \leq 14$.*

Proof. The Fano plane has point set $\mathbb{F}_2^3 \setminus \{0\}$ and lines $\{x, y, x + y\}$ for distinct nonzero vectors x, y. Embed its seven points in the ground set. Every two-coloring of its points has a monochromatic line. Indeed, if there were none, choose three points x, y, z in the larger color class. The three points are noncollinear, and their pairwise line completions $x + y, x + z, y + z$ must lie in the other color. Those three distinct completions sum to zero, so they form a monochromatic line, a contradiction. Every n-set A or its complement contains a Fano line. The seven lines therefore satisfy Lemma 7.2 with $d = 3$, giving fourteen colors. □

Proposition 7.4 (Affine-Walsh coloring)**.** *Let $h \geq 2$ and put $N = 2^{2h}$. Then*

$$\chi\big(\mathrm{KG}(N, N/2, d)\big) \leq 2(N - 1) \qquad \textit{for every } d \leq 2^{h-1} + 2.$$

Consequently,

$$\chi\big(\mathrm{KG}(2n, n, d)\big) \leq 2(2^{2h} - 1)$$

for every $n \geq 2^{2h-1}$ and every $d \leq 2^{h-1} + 2$.

Proof. See Appendix A.1 for the witness construction and its verification. □

Proposition 7.5 (A projective 62-coloring for $d = 5$)**.**

$$\chi\big(\mathrm{KG}(32, 16, 5)\big) \leq 62.$$

Consequently,

$$\chi\big(\mathrm{KG}(2n, n, 5)\big) \leq 62 \qquad (n \geq 16).$$

Proof. See Appendix A.2 for the witness construction and its verification. □

For the comparison below, the complete-family bound uses all $(c + 1)$-subsets of a fixed $(2c + 1)$-set, giving $2\binom{2c+1}{c+1}$ colors by Lemma 7.2.

Corollary 7.6 (Higher EFc counterexamples)**.** *For each row below and every prescribed nonempty two-group split with at least the displayed number of agents, there are monotone valuations on $2r$ goods with no balanced EFc allocation.*

c	*Number of agents*	*Complete-family bound*	*Range*
2	*14*	*20*	$r \geq 4$
3	*30*	*70*	$r \geq 8$
4	*62*	*252*	$r \geq 16$
5	*126*	*924*	$r \geq 32$

Proof. The color bounds are Proposition 7.3, Proposition 7.4 with $h = 2$ and $h = 3$, and Proposition 7.5. For a prescribed split, assign the colors to that many agents using the construction in Theorem 3.3, and give all extra agents zero valuations. If fewer colors are used, assign only the used classes. Every balanced allocation is rejected by its color's agent. □

The finite constructions give smaller color budgets than the complete-family bound in the displayed ranges.

For EF2 this places the universal monotone balanced threshold between five and thirteen agents in the indicated range. Corollary 6.3 reaches six when one valuation is additive. The exact higher thresholds remain open.

A Finite witness colorings

The following constructions supply the upper bounds used in Section 7. Both use the signed discrepancy witness lemma; the balancing transfer extends their finite colorings to the stated ranges.

A.1 The affine-Walsh construction

Proof of Proposition 7.4. Let $V = \mathbb{F}_2^{2h}$, so $|V| = N$. Let V^* be the space of linear maps from V to $\mathbb{F}_2$. For each nonzero functional $a \in V^*$ put $H_a = \{x \in V : a(x) = 0\}$. These $N - 1$ hyperplanes have size $N/2$ and will be our witnesses.

Let $A \subseteq V$ be a half-set, $|A| = N/2$, and define $g_A : V \to \{-1, 1\}$ by

$$g_A(x) = \begin{cases} 1, & x \in A, \\ -1, & x \notin A. \end{cases}$$

Then $\sum_x g_A(x) = 0$. For $a \in V^*$, read $a(x) \in \mathbb{F}_2$ as the integer zero or one, and define the Walsh coefficient

$$W_a(A) := \sum_{x \in V} g_A(x)(-1)^{a(x)}.$$

Orthogonality of the functions $x \mapsto (-1)^{a(x)}$ gives the finite Walsh Parseval identity:

$$\sum_{a \in V^*} W_a(A)^2 = N^2,$$

and $W_0(A) = 0$. Hence some nonzero a satisfies $|W_a(A)| > \sqrt{N} = 2^h$. Since

$$W_a(A) = 2 \sum_{x \in H_a} g_A(x)$$

and $N/2$ is even, each $W_a(A)$ is divisible by 4. Therefore $|W_a(A)| \geq 2^h + 4$. Consequently

$$\left| \sum_{x \in H_a} g_A(x) \right| \geq 2^{h-1} + 2.$$

For one of A, A^c, the sum is positive, which means that its intersection with H_a has size at least $(|H_a| + 2^{h-1} + 2)/2$. Lemma 7.2 therefore gives $2(N - 1)$ colors for $d \leq 2^{h-1} + 2$. The final statement follows from Lemma 7.1. □

A.2 The projective construction

Proof of Proposition 7.5. The binary projective space $\Pi = \mathrm{PG}(4, 2)$ has as its points the nonzero vectors of $\mathbb{F}_2^5$. Its hyperplanes are the nonzero vectors in the kernels of nonzero linear functionals on $\mathbb{F}_2^5$. We also use Π to denote its point set. We use the standard incidence parameters of this projective geometry [8]: Π has 31 points and 31 hyperplanes,

each hyperplane has 15 points, each point lies in 15 hyperplanes, and each pair of points lies in 7 hyperplanes. Add one new point $\infty \notin \Pi$, so the ground set has 32 points.

For each hyperplane $H \subseteq \Pi$, use two colors H^+ and H^-. We first show that every 16-set $A \subseteq \Pi \cup \{\infty\}$ has a hyperplane H such that

$$|A \cap H| \geq 10 \qquad \text{or} \qquad |A \cap H| \leq 5.$$

Put $X = A \cap \Pi$. Since replacing A by its complement interchanges the two alternatives, it is enough to consider the case $|X| = 16$. Suppose, for contradiction, that every hyperplane H satisfies

$$6 \leq |X \cap H| \leq 9.$$

For $j = 6, 7, 8, 9$, let

$$n_j := |\{H : |X \cap H| = j\}|.$$

Counting hyperplanes, point-hyperplane incidences, and ordered pairs of points inside hyperplane intersections gives

$$\begin{aligned} n_6 + n_7 + n_8 + n_9 &= 31, \\ 6n_6 + 7n_7 + 8n_8 + 9n_9 &= 16 \cdot 15 = 240, \\ 36n_6 + 49n_7 + 64n_8 + 81n_9 &= 16 \cdot 15 + 2\binom{16}{2}7 = 1920. \end{aligned}$$

The nonnegative integer solutions are

$$(n_6, n_7, n_8, n_9) = (11, 3, 0, 17) \quad \text{or} \quad (12, 0, 3, 16).$$

Consider the first case. There are three hyperplanes meeting X in exactly seven points. Fix a point $p \in \Pi \setminus X$. The sum of $|X \cap H|$ over the 15 hyperplanes through p is $16 \cdot 7 = 112$. If t_p is the number of the three type-seven hyperplanes through p, and s_p is the number of type-nine hyperplanes through p, then

$$6(15 - t_p - s_p) + 7t_p + 9s_p = 112,$$

so $t_p + 3s_p = 22$. Since $0 \leq t_p \leq 3$, we get $t_p = 1$. Thus every point of $\Pi \setminus X$ lies in exactly one type-seven hyperplane. This gives 15 incidences between outside points and type-seven hyperplanes. But each type-seven hyperplane contains $15 - 7 = 8$ outside points, so the same number of incidences is $3 \cdot 8 = 24$, a contradiction.

In the second case, there are three hyperplanes meeting X in exactly eight points. With t_p the number of type-eight hyperplanes through $p \in \Pi \setminus X$, and s_p the number of type-nine hyperplanes through p, the same local count gives

$$6(15 - t_p - s_p) + 8t_p + 9s_p = 112,$$

so $2t_p + 3s_p = 22$. Since $0 \leq t_p \leq 3$, we get $t_p = 2$. Thus the number of incidences between outside points and type-eight hyperplanes is $15 \cdot 2 = 30$. But the three type-eight hyperplanes contain only $3(15 - 8) = 21$ outside incidences, again a contradiction.

Therefore some hyperplane H satisfies $|A \cap H| \geq 10$ or $|A^c \cap H| \geq 10$. Since $|H| = 15$, the thirty-one hyperplanes satisfy Lemma 7.2 with $d = 5$, giving sixty-two colors.

The large-n statement follows from Lemma 7.1. □

B Exact checks, separate from the proof

The source package contains `verify.py`, `audit_checks.py`, and `joint_checks.py`, together with their recorded JSON outputs. These files use only the Python standard library and are prepared as arXiv ancillary material in `Balanced_Kneser_EFc_Joint_Paper_Package.zip`. They record finite checks of the geometric and coloring constructions. The package includes the expected outputs, `SHA256SUMS`, and a `REPRODUCIBILITY.md` file with exact commands, the software environment, and measured running times and peak memory use. The smaller suite is run by `make check`; `make check-full` runs all reported topological checks sequentially.

The script `verify.py` reconstructs the Johnson-ball complex through dimension four and performs exact Gaussian elimination over $\mathbb{F}_2$. Write f_j for the number of j-dimensional faces and $\beta_j = \dim_{\mathbb{F}_2} H_j(\mathcal{C}(2n, n); \mathbb{F}_2)$ for the jth Betti number. The generator span rank is the dimension of the subspace of third homology spanned by the indicated pentagon cycles. The following exact values are recorded in the supplied certificates:

n	f_0	f_1	f_2	f_3	f_4
3	20	180	960	2,850	4,500
4	70	1,820	22,960	129,500	402,472

n	$(\beta_0, \beta_1, \beta_2, \beta_3)$	all pentagons	one per face	span rank
3	$(1, 0, 0, 11)$	144	12	11
4	$(1, 0, 0, 239)$	4,032	336	239

Here "span rank" is the rank modulo four-simplex boundaries. It is the same for the full family and for the smaller family taking one cyclic order per non-simplex four-face. The latter check is implemented in `audit_checks.py`.

That script also contains a separately written relative cellular calculation for (P, Q). Four-cells are the non-simplex four-faces, and the boundary of each non-simplex five-face is obtained directly by fixing one free coordinate to zero or one. Its chain groups are enumerated directly from the faces of the hypersimplex:

n	$\dim C_4(P, Q)$	$\dim C_5(P, Q)$	$\operatorname{rank} \partial_5$	$\dim H_4(P, Q)$
3	12	1	1	11
4	336	112	97	239
5	5,040	2,940	2,141	2,899
6	55,440	46,200	28,841	26,599

The agreement for $n = 3, 4$ compares two different chain models. The $n = 5, 6$ entries are computations of the polyhedral model, not direct enumerations of the corresponding Johnson complexes.

Finally, all $5^5 = 3,125$ assignments to the chords of a fixed rainbow five-cycle were tested; none satisfies Lemma 2.2's pair-cover condition. For the cap classification, exact rational polygon clipping checks all 42 local patterns in (6) for $\delta = 0, 1, 2$. They yield 14 restriction shapes, of which 12 are nonempty and nontrivial. Among their $2^{12} = 4,096$ subfamilies, exactly one has nonempty intersection but no square corner: the family of all four diagonal half-squares. Larger core deficits are impossible by the free-mass bound in the proof.

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